Optimal strategies in the all-heads coin game

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Abstract

We study a sequential coin-flipping game in which a player starts with n coins, each landing heads independently with probability p. In each round the player flips all remaining coins and must set aside at least one coin showing heads; if no coin shows heads, the player loses. The player wins if and when all coins have been set aside. This is a stochastic shortest-path Markov decision process whose Bellman equation involves a nonlinear suffix-maximum operator. We analyse two natural strategies — One (set aside exactly one head) and All (set aside every head) — and determine the optimal winning probability $w_{n,p}$ as a function of n and p.

For $p = \frac{1}{2}$ every strategy that accepts the immediate win achieves $w_{n,1/2} = \frac{1}{2}$. For $p > \frac{1}{2}$ the strategy One is optimal and $n \mapsto w_{n,p}$ is strictly increasing. For $p < \frac{1}{2}$ near $\frac{1}{2}$ we carry out a first-order perturbation expansion in $\delta := \frac{1}{2} - p$ and prove that the leading coefficient c_n of the deficit $\frac{1}{2} - w_{n,p}$ satisfies a linear recursion for $n \geq 7$, with a limit $L = \lim c_n \approx 1.7035$. As a consequence the optimal value sequence has, to first order in δ, a strict local minimum at $n = 5$ and no local maximum — local maxima are a non-perturbative phenomenon.

1 Introduction

1.1 The game

Fix a probability $p \in (0, 1)$ and a positive integer n. A player is given n coins, each of which independently lands heads with probability p and tails with probability $q := 1-p$. The game proceeds in rounds. In each round:

(i) All remaining coins are flipped simultaneously.

(ii) If no coin shows heads, the game is lost.

(iii) Otherwise the player must set aside at least one coin that shows heads. The set-aside coins leave the game permanently.

(iv) If every coin has been set aside, the game is won. Otherwise, a new round begins with the remaining coins.

The player's objective is to maximise the probability of winning.

Two natural strategies present themselves:

- **Strategy One:** set aside exactly one head when $1 \le k \le n-1$ heads appear, and set aside all n when $k = n$.
- **Strategy All:** set aside every head that appears ($i = k$).

Strategy One is conservative: it preserves as many coins as possible for future rounds, betting that the one set-aside coin has already "done its job." Strategy All is greedy: it cashes in every success immediately. Neither is universally optimal.

A variant of this game, in which the objective is to maximise the expected number of heads collected rather than the probability of collecting all of them, was introduced and studied by van Doorn [6]. The all-heads-wins formulation studied here leads to a richer structural picture.

1.2 Markov decision processes and the Bellman equation

The game is a finite-state *Markov decision process* (MDP): the state is the number m of coins remaining, the action is the number $i \in \{1, \dots, k\}$ of heads to set aside (where k is the random number of heads), and the transition is $m \mapsto m - i$. There are two absorbing states: *win* ($m = 0$, all coins collected) and *lose* (reached when $k = 0$ in some round). A *policy* (or *strategy*) π specifies, for each state n and each observed number $k \in \{1, \dots, n\}$ of heads, the number $i_\pi(n, k) \in \{1, \dots, k\}$ of heads to set aside; we write v_n^π for the resulting winning probability starting from state n.

Because every policy absorbs with probability at least $p^m + (1-p)^m > 0$ per round, the MDP is a *stochastic shortest-path* (SSP) problem in the sense of Bertsekas and Tsitsiklis [2]: the Bellman operator is a contraction and the optimal value function is its unique fixed point.

Definition 1.1 (Optimal winning probability)**.** Let $w_{0,p} := 1$. For $n \ge 1$ define

$$w_{n,p} \;:=\; p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \max_{j \le m \le n-1} w_{m,p}, \tag{1}$$

where $q = 1 - p$ and the empty sum (at $n = 1$) gives $w_{1,p} = p$.

The term p^n accounts for the event that all n coins land heads (an immediate win). Each summand with $j \ge 1$ accounts for the event that exactly j coins land tails; the player then chooses to keep some number $m \in \{j, \dots, n-1\}$ of coins (setting aside $n - m$ heads), and the game continues from state m with optimal value $w_{m,p}$.

The Bellman equation (1) is the standard optimality equation for this SSP–MDP. For general background on Markov decision processes and the Bellman equation we refer to the monographs of Puterman [4], Bertsekas [1], and Stokey, Lucas and Prescott [5].

1.3 Strategies One and All

Under strategy One the player always keeps $m = n - 1$ coins (except when $k = n$), so the Bellman recursion collapses to

$$a_{n,p} \;=\; p^n + \big(1 - p^n - (1-p)^n\big)\, a_{n-1,p}, \qquad a_{0,p} := 1. \tag{2}$$

Under strategy ALL the player always keeps $m = j$ coins (the number of tails), giving

$$b_{n,p} \;=\; p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \, b_{j,p}, \qquad b_{0,p} := 1. \tag{3}$$

Note that the recursions for both a and b are *linear* (no max operator), which makes them analytically more tractable than the Bellman equation (1).

1.4 Outline

In Section 2 we prove that at $p = \frac{1}{2}$ the optimal winning probability is $w_{n,1/2} = \frac{1}{2}$ for every n, and that in fact every reasonable strategy achieves this value. In Section 3 we show that for $p > \frac{1}{2}$ the strategy ONE is always optimal and $n \mapsto w_{n,p}$ is strictly increasing. Section 4 treats the case $p < \frac{1}{2}$ near $\frac{1}{2}$: we introduce a first-order perturbation in $\delta = \frac{1}{2} - p$, derive a recursion for the leading coefficient c_n, prove that it simplifies to a linear recursion for $n \geq 7$, and compute the limit $L = \lim c_n$.

2 The fair coin: $p = \frac{1}{2}$

Theorem 2.1. *For every $n \geq 1$, $w_{n,1/2} = \frac{1}{2}$. More generally, for any policy π satisfying $i_\pi(n,n) = n$ for every n (i.e. the player accepts the immediate win when all coins land heads), the winning probability is $v_n^\pi = \frac{1}{2}$.*

In particular, both strategies ONE and ALL achieve the optimum: $a_{n,1/2} = b_{n,1/2} = w_{n,1/2} = \frac{1}{2}$.

Proof. We use the binomial identity

$$\sum_{j=1}^{n-1} \binom{n}{j} \left(\frac{1}{2}\right)^n \;=\; 1 - 2^{1-n}. \tag{4}$$

Step 1. We show $w_{n,1/2} = \frac{1}{2}$ by strong induction. For $n = 1$: $w_{1,1/2} = \frac{1}{2}$. For $n \geq 2$, suppose $w_{m,1/2} = \frac{1}{2}$ for all $1 \leq m \leq n-1$. Then every suffix-maximum equals $\frac{1}{2}$, so

$$\begin{aligned} w_{n,1/2} &= 2^{-n} + \frac{1}{2} \sum_{j=1}^{n-1} \binom{n}{j} 2^{-n} \\ &= 2^{-n} + \tfrac{1}{2}\big(1 - 2^{1-n}\big) \\ &= 2^{-n} + \tfrac{1}{2} - 2^{-n} \;=\; \tfrac{1}{2}. \end{aligned}$$

Step 2. For a general policy π with $i_\pi(n,n) = n$, conditioning on the number of heads $K \sim \mathrm{Bin}(n, \frac{1}{2})$ gives

$$v_n^\pi = 2^{-n} \cdot 1 + \sum_{k=1}^{n-1} \binom{n}{k} 2^{-n} \, v^\pi_{n - i_\pi(n,k)}.$$

Since $1 \leq i_\pi(n,k) \leq k \leq n-1$ for $k < n$, the subscript $n - i_\pi(n,k)$ lies in $\{1, \ldots, n-1\}$, and by strong induction each $v^\pi_{n-i_\pi(n,k)} = \frac{1}{2}$. The same calculation as in Step 1 gives $v_n^\pi = \frac{1}{2}$. $\square$

Remark 2.2. A policy that refuses the immediate win at $k = n$ (i.e. $i_\pi(n, n) < n$) achieves strictly less than $\frac{1}{2}$: the contribution of the $k = n$ event drops from 2^{-n} to $2^{-n} \cdot v^\pi_{n-i_\pi(n,n)} < 2^{-n}$, while all other terms remain at most $\frac{1}{2} \cdot \binom{n}{k} 2^{-n}$, giving $v^\pi_n < \frac{1}{2}$.

3 Above the fair coin: $p > \frac{1}{2}$

Theorem 3.1. *For $p > \frac{1}{2}$ and all $n \geq 2$:*

(i) $w_{n,p} > w_{n-1,p}$, *i.e. $n \mapsto w_{n,p}$ is strictly increasing.*

(ii) $w_{n-1,p} < \dfrac{p^n}{p^n + q^n}$.

(iii) *Strategy* One *is optimal:* $w_{n,p} = a_{n,p}$.

Proof. We prove (i) and (ii) simultaneously by strong induction on n.

Base case $n = 2$. One computes $w_{1,p} = p$ and $w_{2,p} = p^2 + (1 - p^2 - q^2)p = 3p^2 - 2p^3$. Then $w_{2,p} - w_{1,p} = p(1 - p)(2p - 1) > 0$ since $p > \frac{1}{2}$. Also $w_{1,p} = p < p^2/(p^2 + q^2)$ iff $p^2 + q^2 > p$, which holds for $p \in (\frac{1}{2}, 1)$.

Inductive step. Assume (i) and (ii) hold for all indices up to $n - 1$. Since $n \mapsto w_{n,p}$ is increasing on $\{1, \ldots, n - 1\}$ by hypothesis, the suffix-maximum $\max_{j \leq m \leq n-1} w_{m,p} = w_{n-1,p}$ for every $j \in \{1, \ldots, n - 1\}$. Hence the Bellman equation (1) reduces to

$$w_{n,p} = p^n + (1 - p^n - q^n)\, w_{n-1,p},$$

which is exactly the recursion (2) for strategy One. This proves (iii).

For (i), $w_{n,p} > w_{n-1,p}$ iff $p^n(1 - w_{n-1,p}) > q^n w_{n-1,p}$, i.e. $w_{n-1,p} < p^n/(p^n + q^n)$, which is hypothesis (ii) at level $n - 1$.

For (ii) at level n, we need $w_{n,p} < p^{n+1}/(p^{n+1} + q^{n+1})$. Using the recursion and the inductive bound on $w_{n-1,p}$, a direct computation shows this reduces to the inequality $(q/p)^n < (q/p)^{n-1}$, which holds since $q/p < 1$. □

4 Below the fair coin: perturbation in $\delta = \frac{1}{2} - p$

For $p < \frac{1}{2}$ the structure of $n \mapsto w_{n,p}$ is far richer: numerical experiments show that the sequence develops local minima and maxima, and the optimal strategy is neither One nor All in general; see [6], Section 10.

To get analytical traction, we study the regime $p = \frac{1}{2} - \delta$ with $\delta > 0$ small.

4.1 The deficit and its recursion

Definition 4.1. For $p \in (0, 1)$ and $n \geq 1$, define the *deficit* $\Delta_{n,p} := \frac{1}{2} - w_{n,p}$.

By Theorem 2.1, $\Delta_{n,1/2} = 0$ for all n.

Proposition 4.2. *The deficit satisfies*

$$\Delta_{n,p} \;=\; \frac{q^n - p^n}{2} + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \min_{j \le m \le n-1} \Delta_{m,p}, \tag{5}$$

with $\Delta_{0,p} = -\frac{1}{2}$. *In particular:*

(i) *For* $p < \frac{1}{2}$: $\Delta_{n,p} > 0$ *for every* $n \ge 1$ *(i.e.* $w_{n,p} < \frac{1}{2}$*).*

(ii) *For* $p > \frac{1}{2}$: $\Delta_{n,p} < 0$ *for every* $n \ge 1$ *(i.e.* $w_{n,p} > \frac{1}{2}$*).*

Proof. Substitute $w_{m,p} = \frac{1}{2} - \Delta_{m,p}$ into the Bellman equation (1). The max on w_m becomes a min on Δ_m. The constant terms $\frac{1}{2} \cdot \big[\text{binomial weights}\big]$ cancel by the identity $p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \cdot \frac{1}{2} = \frac{1}{2} - \frac{1}{2}(q^n - p^n)$, leaving (5). The sign claims follow by induction using $q^n - p^n > 0$ (resp. < 0) and the positivity (resp. negativity) of the min terms. □

4.2 First-order expansion

Set $p = \frac{1}{2} - \delta$ with $\delta > 0$ small. Then $q = \frac{1}{2} + \delta$, and the constant term in (5) is

$$\frac{q^n - p^n}{2} = \frac{(\frac{1}{2} + \delta)^n - (\frac{1}{2} - \delta)^n}{2} = \frac{n\delta}{2^{n-1}} + O(\delta^3).$$

The binomial weights become $\binom{n}{j} 2^{-n} + O(\delta)$, and the min terms are $O(\delta)$ by Proposition 4.2. Thus $\Delta_{n,p} = O(\delta)$.

Proposition 4.3 (First-order coefficient)**.** *Write* $\Delta_{n,p} = c_n \delta + O(\delta^2)$ *as* $\delta \to 0^+$. *Then* $c_1 = 1$ *and for* $n \ge 2$*:*

$$c_n \;=\; \frac{n}{2^{n-1}} + \frac{1}{2^n} \sum_{j=1}^{n-1} \binom{n}{j} \min_{j \le m \le n-1} c_m. \tag{6}$$

Proof. Match first-order coefficients in (5). The constant term contributes $n/2^{n-1}$. Each binomial weight is $\binom{n}{j} 2^{-n} + O(\delta)$; the min of the $\Delta_m = c_m \delta + O(\delta^2)$ at first order is $(\min c_m)\delta + O(\delta^2)$. Products of $O(\delta)$ terms are $O(\delta^2)$. □

Example 4.4. The first few values of c_n:

$$c_1 = 1, \quad c_2 = \tfrac{3}{2}, \quad c_3 = \tfrac{27}{16}, \quad c_4 = \tfrac{111}{64}, \quad c_5 = \tfrac{3555}{2048} \approx 1.7358, \quad c_6 = \tfrac{113337}{65536} \approx 1.7293.$$

The sequence increases on $\{1, \ldots, 5\}$ and decreases for $n \ge 6$. In particular $c_4 = \frac{111}{64} > c_6$ and $c_5 > c_6$, both of which we use below as base cases.

4.3 Collapse, lower bound, and linear recursion

The four results below — the asymptotic collapse of the min in (6) (Lemma 4.5), the uniform lower bound $c_m \geq \frac{27}{16}$ (Lemma 4.6), the strict decrease of $(c_n)_{n\geq 5}$ (Lemma 4.7), and the linear recursion for $n \geq 7$ (Proposition 4.8) — are mutually entangled: each, taken alone, depends on parts of the others. Stating them in isolation produces an apparent circular dependency. We therefore state all four results first and then prove them simultaneously by induction on n, with the IH at level $< n$ supplying exactly what is needed at level n.

Lemma 4.5 (Asymptotic collapse)**.** *For all $n \geq 7$ and $j \in \{1, \ldots, n-1\}$:*

$$\min_{j\leq m\leq n-1} c_m \;=\; \begin{cases} c_j & \text{if } j \in \{1,2,3\}, \\ c_{n-1} & \text{if } j \in \{4,\ldots,n-1\}. \end{cases} \tag{7}$$

Lemma 4.6. *$c_m \geq \frac{27}{16}$ for every $m \geq 4$.*

Lemma 4.7. *The sequence $(c_n)_{n\geq 5}$ is strictly decreasing.*

Proposition 4.8 (Linear recursion)**.** *For $n \geq 7$:*

$$c_n \;=\; A_n + (1 - B_n)\, c_{n-1}, \tag{8}$$

where

$$A_n := \frac{n}{2^{n-1}} + \frac{n\,c_1 + \binom{n}{2}c_2 + \binom{n}{3}c_3}{2^n}, \qquad B_n := \frac{2 + n + \binom{n}{2} + \binom{n}{3}}{2^n}. \tag{9}$$

Moreover the algebraic identity

$$A_n - \tfrac{27}{16}B_n \;=\; -\frac{3(n^2 - 15n + 36)}{32 \cdot 2^n} \tag{10}$$

holds for every $n \geq 1$ (independently of the linear recursion).

Joint proof of Lemmas 4.5, 4.6, 4.7 and Proposition 4.8. The identity (8) is a direct algebraic computation from (10) using $\binom{n}{2} = \frac{n(n-1)}{2}$ and $\binom{n}{3} = \frac{n(n-1)(n-2)}{6}$; we use it freely below.

For $n \geq 4$ let $P(n)$ denote the conjunction of the following four claims (each understood vacuously when its index range is empty):

(a) Lemma 4.5 holds at level n (when $n \geq 7$);

(b) $c_n \geq \frac{27}{16}$;

(c) $c_n < c_{n-1}$ (when $n \geq 6$);

(d) the linear recursion (9) holds at level n (when $n \geq 7$).

We prove $P(n)$ for all $n \geq 4$ by strong induction on n.

Base cases $n \in \{4, 5, 6\}$. By Example 4.4, $c_4 = \frac{111}{64}$, $c_5 = \frac{3555}{2048}$, $c_6 = \frac{113337}{65536}$, all of which exceed $\frac{27}{16} = \frac{108}{64}$, so (b) holds. Also $c_5 > c_6$, so (c) holds at $n = 6$. Claims (a) and (d) are vacuous for $n \leq 6$.

Inductive step. Fix $n \geq 7$ and assume $P(m)$ for all $4 \leq m < n$. In particular: by IH(b), $c_m \geq \frac{27}{16}$ for $4 \leq m \leq n-1$; by IH(c), $c_5 > c_6 > \cdots > c_{n-1}$. We now establish (a)–(d) at level n.

(a) Collapse. For $j \in \{1,2,3\}$: from $c_1 = 1 < c_2 = \frac{3}{2} < c_3 = \frac{27}{16}$ (Example 4.4) and $c_m \geq \frac{27}{16} = c_3$ for $m \geq 4$ (IH(b)), the minimum over $\{j, \ldots, n-1\}$ is c_j.

For $j \in \{4, \ldots, n-1\}$: by IH(c), $(c_m)_{m\geq 5}$ is strictly decreasing, so $c_5 > c_6 > \cdots > c_{n-1}$, and in particular $c_m \geq c_{n-1}$ for $m \in \{5, \ldots, n-1\}$. For $m = 4$, $c_4 = \frac{111}{64} > c_6$ (Example 4.4) and $c_6 \geq c_{n-1}$ (IH(c)), so $c_4 \geq c_{n-1}$ as well. Hence the minimum over $\{j, \ldots, n-1\}$ for $j \geq 4$ is c_{n-1}.

(d) Linear recursion. Substituting the collapse (a) just proved into (6) and splitting the sum at $j = 3$ yields (9)–(10).

(b) Lower bound. From (d) and (8),

$$c_n - \tfrac{27}{16} \;=\; \big(A_n - \tfrac{27}{16}B_n\big) + (1 - B_n)\big(c_{n-1} - \tfrac{27}{16}\big).$$

By IH(b), $c_{n-1} - \frac{27}{16} \geq 0$; one checks $0 < B_n < 1$ for all $n \geq 7$. For $n \in \{7, \ldots, 12\}$, the quadratic $n^2 - 15n + 36$ is non-positive on $\{3, \ldots, 12\}$ (its roots are 3 and 12), so $A_n - \frac{27}{16}B_n \geq 0$ by (8), and (b) follows. For $n \geq 13$ we argue cumulatively: define $\varepsilon_m := c_m - \frac{27}{16}$, so (d) gives $\varepsilon_m = (1 - B_m)\varepsilon_{m-1} + (A_m - \frac{27}{16}B_m)$. The starting value $\varepsilon_{12} > \frac{1}{60}$ is verified by exact rational computation; the cumulative negative contribution is bounded by $\sum_{k\geq 13} \frac{3(k^2-15k+36)}{32\cdot 2^k} \leq \frac{297}{65536} < \frac{1}{200}$ (using $\sum_{k\geq 13} k^2/2^k = \frac{99}{2048}$); meanwhile $\prod_{m\geq 13}(1 - B_m) > \frac{7}{8}$ (using $\sum_{m\geq 13} B_m < \frac{1}{8}$). So $\varepsilon_n > \frac{1}{60} \cdot \frac{7}{8} - \frac{1}{200} = \frac{23}{2400} > 0$ for $n \geq 13$.

(c) Strict decrease. By (d), $c_n - c_{n-1} = A_n - B_n c_{n-1}$. Using $c_{n-1} \geq \frac{27}{16}$ (IH(b)) and $B_n > 0$, this is $\leq A_n - \frac{27}{16}B_n < 0$ for $n \geq 13$ by (8) ($n^2 - 15n + 36 > 0$ there). For $n \in \{7, \ldots, 12\}$, $c_n < c_{n-1}$ is verified by direct rational computation. This completes the inductive step. □

4.4 The limit

Theorem 4.9 (Existence and value of the limit)**.** *The limit $L := \lim_{n\to\infty} c_n$ exists, and for any $n_0 \geq 7$:*

$$L \;=\; c_{n_0-1} \prod_{m\geq n_0} (1 - B_m) \;+\; \sum_{k\geq n_0} A_k \prod_{m>k} (1 - B_m). \tag{11}$$

Both the infinite product and the series converge at geometric rate since $A_n, B_n = O(n^3/2^n)$. Numerically,

$$L \;=\; 1.70347176087173673645\ldots \tag{12}$$

Proof. From (9), $c_n = A_n + (1 - B_n)c_{n-1}$. Since $B_n > 0$ and $A_n > 0$ for all n, and (c_n) is eventually decreasing and bounded below (by Lemma 4.6), the limit exists. Iterating the recursion from $n = n_0$ and taking $n \to \infty$ yields (11); the rate follows from $B_n = O(n^3/2^n)$. □

4.5 Shape of the optimal-value sequence near $p = \frac{1}{2}$

Corollary 4.10. *For $\delta > 0$ sufficiently small and $p = \frac{1}{2} - \delta$:*

(i) $w_{n-1,p} - w_{n,p} = (c_n - c_{n-1})\delta + O(\delta^2)$.

(ii) *The sequence $n \mapsto w_{n,p}$ has, to first order in δ, a strict local minimum at $n = 5$ (since $c_5 > c_4$ and $c_5 > c_6$).*

(iii) *There is no local maximum at first order: c_n is eventually decreasing, so $w_{n,p}$ is eventually increasing.*

Remark 4.11. Numerical computation shows that the sequence $n \mapsto w_{n,p}$ does develop strict local maxima for p bounded away from $\frac{1}{2}$ (e.g. at $n = 9$ for $p = 0.42$, verified by exact rational arithmetic). This is consistent with the corollary: local maxima are a *non-perturbative* phenomenon invisible to the first-order expansion. Van Doorn [6], Section 10, conjectures that the optimal strategy for $p < \frac{1}{2}$ is "infinitely complex."

References

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